SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – EXPERIMENTATION

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – EXPERIMENTATION
Weightage:	30 % of CIA	**Total Marks:**	100 marks
CO2 Statement:	*Compare mobile communication technologies, including multiple access techniques and network evolution, based on performance and applications.*		
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Section / Batch:	CSE-AI	**Date:**	11/10/2026

Code of Conduct

*I **Shaik Azeem(192472307)**(Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Shaik Azeem

Instructions

- This test contains FIVE questions.
- Duration: 60 minutes (homework)

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Annexure A - QUESTIONS

1. Figma-Based Mobile IP Communication Workflow Analysis (BL4 – Analyze)

Design an interactive Figma prototype illustrating the complete workflow of Mobile IP communication when a Mobile Node moves from its Home Network to a Foreign Network. Your design should include the Home Agent (HA), Foreign Agent (FA), Correspondent Node (CN), Care-of Address (CoA), Registration Process, and Packet Tunneling.

Analyze the communication process by identifying:

- The purpose of each network component.
- Points where packet delay or packet loss may occur.
- The effect of node mobility on communication performance.
- Possible improvements to reduce latency.

Explanation

Mobile IP allows a device to move between IP networks while maintaining its permanent home IP address for communication. It helps other devices continue reaching the mobile device even when its current network changes.

Architecture

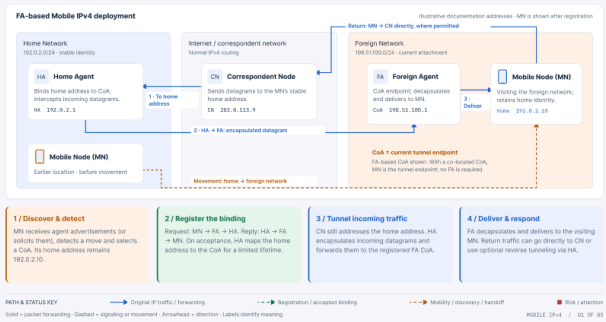
E1 MOBILE COMPUTING LABORATORY / EXPERIMENT 1

CONCEPTUAL DEMONSTRATION

Mobile IP Communication Workflow

Architecture 01

Aim: preserve a Mobile Node's home address while it moves from its Home Network to a Foreign Network.



1 / Discover & detect
MN receives agent advertisements (or solicits them), detects a move and selects a CoA. Its home address remains 192.0.2.10.

2 / Register the binding
Request: MN → FA → HA. Reply: HA → FA → MN. On acceptance, HA maps the home address to the CoA for a limited lifetime.

3 / Tunnel incoming traffic
CN still addresses the home address. HA encapsulates incoming datagrams and forwards them to the registered FA CoA.

4 / Deliver & respond
FA decapsulates and delivers to the visiting MN. Return traffic can go directly to CN or use optional reverse tunneling via HA.

PATH & STATUS KEY — Original IP traffic / forwarding · Registration / accepted binding · Mobility / discovery / handoff · Risk / attention

Solid = packet forwarding · Dashed = signaling or movement · Arrowhead = direction · Labels identify meaning

MOBILE IPv4 / 01 OF 05

Registration

E1 MOBILE COMPUTING LABORATORY / EXPERIMENT 1

CONCEPTUAL DEMONSTRATION

Registration

02

Agent discovery comes first. A successful exchange installs a time-limited home-address → Care-of Address binding.



Request contents
Home address, HA address, selected CoA, requested lifetime and identification. Authentication extensions protect the registration exchange.

Lifetime & renewal
The HA grants a bounded lifetime in the reply. MN renews before expiry to keep the binding active; expiry removes the forwarding state.

Acceptance is the state change
Movement alone does not install a new binding. A rejected request does not create the requested mapping; a stale binding may still point to the prior CoA.

PATH & STATUS KEY — Original IP traffic / forwarding · Registration / accepted binding · Mobility / discovery / handoff · Risk / attention

Solid = packet forwarding · Dashed = signaling or movement · Arrowhead = direction · Labels identify meaning

MOBILE IPv4 / 02 OF 05

Packet Tunneling

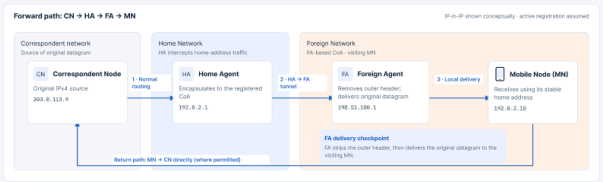
E1 MOBILE COMPUTING LABORATORY / EXPERIMENT 1

CONCEPTUAL DEMONSTRATION

Packet Tunneling

03

Incoming datagrams keep the MN's home address as their inner destination; an outer IPv4 header carries them to the FA CoA.



Before encapsulation
INNER HEADER / CN → MN HOME ADDRESS
src 203.0.113.9 → dst 192.0.2.10
Original payload · unchanged destination identity
CN uses ordinary IPv4 routing. The home address does not reveal the MN's current foreign network.

Inside the HA → FA tunnel
OUTER HEADER / HA → FA CoA
src 192.0.2.1 → dst 198.51.100.1
INNER HEADER / CN → MN HOME ADDRESS
src 203.0.113.9 → dst 192.0.2.10
Original payload · unchanged destination identity

Optional reverse tunneling
Direct return traffic uses the MN home address as its source. If supported and negotiated, reverse tunneling carries return datagrams via FA → HA, then HA routes them to CN. Routing policy or ingress filtering may constrain the direct path.

PATH & STATUS KEY — Original IP traffic / forwarding · Registration / accepted binding · Mobility / discovery / handoff · Risk / attention

Solid = packet forwarding · Dashed = signaling or movement · Arrowhead = direction · Labels identify meaning

MOBILE IPv4 / 03 OF 05

Handoff

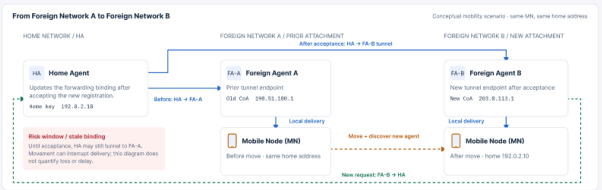
E1 MOBILE COMPUTING LABORATORY / EXPERIMENT 1

CONCEPTUAL DEMONSTRATION

Handoff

04

A second move illustrates the same mechanism: attachment changes first; the HA's forwarding binding changes only after accepted registration.



1 / Move & detect
MN leaves A and detects B through agent discovery. The physical or link-layer move is not itself a new HA binding.

2 / Select & request
MN selects FA-B's CoA and sends a new Registration Request: MN → FA-B → HA. The home address stays unchanged.

3 / Accept & install
HA validates the request and installs the new CoA with a granted lifetime. Registration Reply returns HA → FA-B → MN.

4 / Resume & renew
Incoming datagrams now tunnel to FA-B, which delivers to MN. MN renews before expiry; recovery during movement is not assumed lossless.

PATH & STATUS KEY — Original IP traffic / forwarding · Registration / accepted binding · Mobility / discovery / handoff · Risk / attention

Solid = packet forwarding · Dashed = signaling or movement · Arrowhead = direction · Labels identify meaning

MOBILE IPv4 / 04 OF 05

Analysis

E1 MOBILE COMPUTING LABORATORY / EXPERIMENT 1

CONCEPTUAL DEMONSTRATION

Analysis

05

Reason about identity, binding state and delivery. These scenarios are conceptual demonstrations, not real measurements.

Expected behavior & inspection prompts

Qualitative checks for a lab walkthrough · no latency, throughput or loss values asserted

SCENARIO / STATE	EXPECTED MOBILE IPv4 BEHAVIOR	WHAT TO CHECK
At home HOME ROUTE	With foreign registration removed, MN receives through its home network. No foreign-agent tunnel is needed.	Confirm the MN home address is unchanged and that delivery is local to the home network.
Registered abroad ACCEPTED	HA has an active home → CoA binding. Incoming traffic tunnels to FA, then reaches the visiting MN.	Check accepted reply, granted lifetime, outer destination = FA CoA and inner destination = MN home address.
Move pending IN TRANSITION	MN has moved, but HA may still hold the prior CoA until the new registration is accepted.	Separate discovery, request, acceptance and resumed delivery. Identify the stale-binding risk window.
Expiry or rejection ATTENTION	An expired binding is removed. A rejected request does not install the requested CoA mapping.	Check reply outcome and lifetime state. Do not assume foreign delivery continues without a valid binding.

Risks at the point of impact

Registration security / HA
Authenticate registration and protect against replay; a forged CoA binding could redirect incoming traffic.

Stale binding / prior FA
HA may target an old FA until a new registration succeeds. Binding expiry also ends tunnel forwarding.

Movement loss / MN
Link changes and pending registration can interrupt delivery. This conceptual flow makes no loss or delay guarantee.

HA **Home Agent**
Binding authority and tunnel source

Tunnel

FA **Foreign Agent**
Current CoA and delivery state

Delivery

Mobile Node (MN)
Attachment may change during delivery

Lab conclusion

Mobile IPv4 separates stable identity (home address) from current reachability (CoA). Accepted, renewed bindings let HA direct traffic to the current attachment; movement alone does not update that state.

PATH & STATUS KEY — Original IP traffic / forwarding · Registration / accepted binding · Mobility / discovery / handoff · Risk / attention

Solid = packet forwarding · Dashed = signaling or movement · Arrowhead = direction · Labels identify meaning

MOBILE IPv4 / 05 OF 05

2. Comparative Network Architecture Design Using Figma (BL4 – Analyze)

Using Figma or draw.io, create a comparative architectural diagram showing both a Cellular Wireless Network and a Mobile Ad Hoc Network (MANET) operating in the same smart city environment. Analyze the architectures by comparing:

- Infrastructure dependency.
- Routing mechanisms.
- Scalability.
- Fault tolerance.
- Mobility support.
- Suitable application scenarios.

Highlight the advantages and limitations of each architecture through annotations in your design.

Explanation

A Cellular Network uses infrastructure such as base stations and a cellular core network to provide wireless connectivity.

A MANET is a self-organizing wireless network where mobile devices communicate directly or forward packets through other devices without requiring fixed access infrastructure.

Comparative Network Architecture: Cellular Wireless Network vs MANET

— Architecture

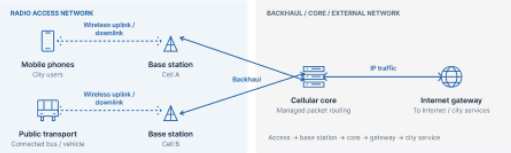
Objective

Compare infrastructure-based cellular communication with mobile ad hoc networking (MANET), focusing on packet paths, mobility and failure behavior in a smart city.

Section 1 · Cellular Network

INFRASTRUCTURE-MEDIATED

Smart city devices reach services through base stations and a managed core. Dashed links are radio access; solid links are backhaul and core IP transport.

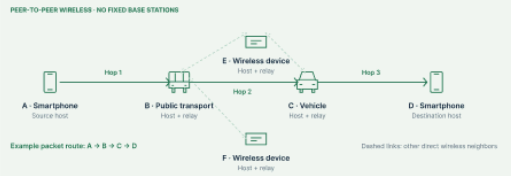


CAUTION · Redundancy can reduce failures, but a base station, backhaul or core outage may interrupt service in the affected area.

Section 2 · MANET

DISTRIBUTED / AD HOC

Nearby nodes communicate directly. For distant peers, intermediate nodes forward packets; each node can act as both a host and a relay.



CAUTION · Mobility changes topology and can break links. Route repair works only if an alternate connected path exists; partitions interrupt communication.

Use-case lens

Smart city communication

Cellular connects city-wide users to services; MANET supports local peer exchanges within a connected group.

Public transport

Cellular provides fleet connectivity with coverage; MANET can share nearby vehicle alerts through relays.

Disaster recovery

MANET can enable local responder messaging without fixed access, but range and partitions limit reach.

Procedure · Conceptual comparison

01 Inspect both architectures and identify access, core and peer nodes.

02 Trace a cellular packet path and the MANET route A → B → C → D.

03 Consider mobility, then a base-station outage or a relay/link failure.

04 Compare routing, reachability and trade-offs using the companion table.

Comparison Table

Comparative Network Architecture: Cellular Wireless Network vs MANET

— Comparison Table

Compare behavior, not absolute winners

Qualitative expectations for the two architectures. Actual performance depends on deployment, radio conditions, traffic and mobility.

Criterion	Cellular Network	MANET
Infrastructure dependency	Requires base stations, backhaul and a managed cellular core; Internet access uses a gateway.	No fixed base stations are required. Communication depends on available peers forming a connected path.
Routing mechanisms	Infrastructure-managed core routing carries traffic; network procedures manage access and handover.	Distributed multi-hop routing, e.g. AODV (on-demand) or OLSR (proactive), selects paths through peer relays.
Scalability	Scales through coverage and capacity planning, spectrum allocation and added sites; congestion still limits capacity.	Routing overhead, contention and interference generally grow with network size or density; performance depends on topology and protocol.
Fault tolerance	Redundant infrastructure can improve availability. Base station, backhaul or core outages can still disrupt service.	May reroute if an alternate connected path exists. Relay loss or link breaks can partition the network; connectivity is not guaranteed.
Mobility support	Mobility is managed through handover between cells, subject to coverage, load and network configuration.	Movement changes neighbor relationships and routes. Route repair or discovery is needed when active links break.
Energy consumption	Device energy depends on signal strength, radio state, traffic and handovers; infrastructure also consumes power.	Device energy depends on traffic and protocol. Relaying, route discovery and listening consume participating nodes' batteries.
Advantages and limitations	Managed wide-area coverage and service capacity. Requires deployed infrastructure, operating resources and available coverage.	Rapid, infrastructure-independent local networking. Limited by radio range, node participation, battery and changing topology.
Suitable applications	City-wide Internet access, public transport fleet telemetry, mobile services and wide-area IoT where coverage is available.	Local responder communication, temporary field networks and nearby vehicle/device cooperation where peers remain connected.

Analysis

Cellular · Planned coverage & capacity

A managed network coordinates access, routing and handover across a coverage area. It suits wide-area services, but depends on functioning infrastructure and sufficient capacity.

MANET · Infrastructure-independent peers

Peer relays support local communication without fixed access. This offers conditional resilience: enough powered nodes and an unbroken connected path must remain available.

Conclusion

Choose cellular for managed wide-area coverage and capacity; choose MANET for temporary or infrastructure-independent local communication. Select by required reach, mobility, available infrastructure, failure risk and energy budget—not by assuming either architecture guarantees connectivity.

Comparative Network Architecture: Cellular Wireless Network vs MANET

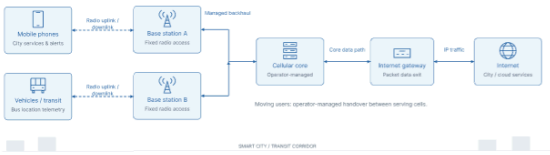
Architecture / Smart city communication models

01 Architecture | 02 Comparison & notes

LINK KEY — Dashed = wireless radio | Solid = managed backhaul / backbone | ↔ Two-way link | Bold → Example packet route

01 Cellular Network

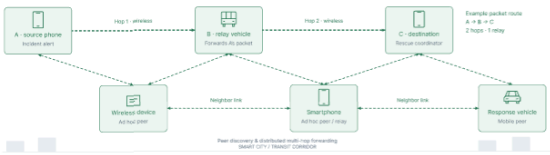
Infrastructure-based communication: devices reach external services through operator-managed access and a central core.



⚠ Infrastructure outage dependence: loss of a serving site, backhaul or core can interrupt service; redundancy may reduce, but not eliminate, this risk.

02 MANET · Mobile Ad hoc Network

Infrastructure-independent peers form direct wireless links. Nodes can forward packets; no fixed base station or central core is required.



⚠ Changing routes & limited battery: movement may break links; discovery and relaying use energy. Alternate routes exist only while peers remain connected.

Smart city · Cellular
Street sensors and citizens' phones send readings or service requests to city platforms through the operator network.

Public transport · Cellular
Buses upload GPS locations and passengers receive live arrival updates; managed handover supports moving users.

Disaster recovery · MANET
Nearby rescue phones and vehicles relay local alerts when infrastructure is unavailable, provided a connected path exists.

Comparative Network Architecture: Cellular Wireless Network vs MANET

Comparison table / Conceptual experiment notes

01 Architecture | 02 Comparison & notes

Objective Compare the architecture, routing, mobility, resilience and energy trade-offs of Cellular Wireless Networks and MANETs, and identify suitable roles in smart cities, public transport and disaster recovery.

Comparison criterion	Cellular Wireless Network	MANET
Infrastructure dependency	Requires operator-managed base stations, access/backhaul and a core network. Internet services use a gateway.	No fixed base station or required central core. Peers form an ad hoc network; external Internet access needs a separate gateway.
Routing mechanisms	Operator-managed access and backbone routing carry traffic through the core; managed handover maintains access as users move.	Distributed ad hoc routing: AODV is reactive; OLSR is proactive. Nodes forward traffic over multi-hop wireless paths.
Scalability	Broad managed coverage and capacity planning support large deployments, but congestion, spectrum and cell capacity constrain service.	Routing overhead, shared-channel interference and hop count limit scale. Performance depends on node density, traffic and mobility.
Fault tolerance	Redundant sites and network paths can improve resilience. Serving-site, backhaul or core outages may still disrupt service.	Alternate paths can bypass a failed node only when a connected route exists. Sparse or separated groups may remain partitioned.
Mobility support	Managed handover between cells supports moving phones and vehicles; coverage gaps or failed handovers can interrupt sessions.	Topology changes as peers move. Routes must be discovered or updated; frequent link breaks can increase delay and packet loss.
Energy consumption	Device power control and network scheduling manage radio energy. Weak signals, mobility and heavy traffic still increase battery use.	Peers spend energy on route discovery, control traffic and relaying others' packets. Relay load can drain some batteries faster.
Advantages and limitations	Established coverage, managed mobility and access to city/cloud services; depends on deployed infrastructure, capacity and operator service.	Rapid local deployment without fixed infrastructure; limited by connectivity, changing routes, interference and available battery power.
Suitable applications	Smart-city sensor uploads, mobile service requests, bus GPS tracking and live passenger information where coverage is available.	Emergency teams, disaster-recovery alerts and temporary vehicle/phone coordination where infrastructure is absent or damaged.

Procedure

1. **Define the two models.**
   Identify fixed infrastructure in cellular and peer nodes in MANET. Assume compatible radios and authorized participation.
2. **Trace a message path.**
   Follow device → base station → core → gateway → Internet; then follow A → B → C in the MANET schematic.
3. **Apply qualitative scenarios.**
   Consider a moving bus, an infrastructure outage, a broken peer link and a low-battery relay. Note when a route remains possible.
4. **Compare and interpret.**
   Use the eight criteria above. Relate strengths and limitations to city services, public transport and disaster recovery.

Qualitative analysis

Normal city & transport operation
Cellular suits bus tracking and cloud-connected services when coverage and capacity are adequate. MANET is useful for nearby peer exchange, not a substitute for wide-area coverage.

Emergency & disaster recovery
A MANET can carry local rescue alerts without working cellular sites. It cannot bridge a partition without a connecting peer or provide Internet access without a gateway.

Movement & battery constraints
Cellular manages handover and device transmit power. MANET must maintain changing routes and share forwarding work; mobile or depleted relays can remove a path.

Conceptual study only. No simulation, timing, throughput or battery measurements are reported.

Conclusion · Choose for the application

There is no universal winner. Cellular is appropriate for managed, wide-area city and transport services where infrastructure is available. MANET is appropriate for temporary, local emergency communication when connected peers can relay traffic. Choose according to coverage, mobility, resilience, capacity and battery needs; hybrid deployments may combine both approaches.

3. Routing Protocol Visualization and Analysis (BL4 – Analyze)

Develop an interactive visualization using Figma, Mermaid.js, D3.js, or any visualization framework illustrating the operation of one Proactive Routing Protocol and one Reactive Routing Protocol in a mobile wireless network.

Analyze:

- Route discovery process.
- Route maintenance mechanism.
- Routing overhead.
- Delay in establishing communication.
- Performance under high node mobility.

Include a comparative summary explaining which protocol performs better under different mobility conditions.

Explanation

Routing protocols determine how packets travel from a source node to a destination node through a network.

For this experiment, your web application simulates a network of nodes, highlights packet paths and demonstrates what happens when wireless links fail.

1. OLSR — Proactive routing

OLSR maintains routing information before a node needs to send data.

Working process:

1. Nodes periodically exchange HELLO messages to discover their neighbors.
2. Nodes exchange topology information.
3. Routing tables are updated.
4. A source sends a packet using an available route.
5. If the topology changes, the routing information is updated.

Advantage: A route may already be available when communication starts.

Limitation: Periodic control messages consume bandwidth, even when some routes are not being used.

2. AODV — Reactive routing

AODV discovers a route when the source needs to communicate with a destination.

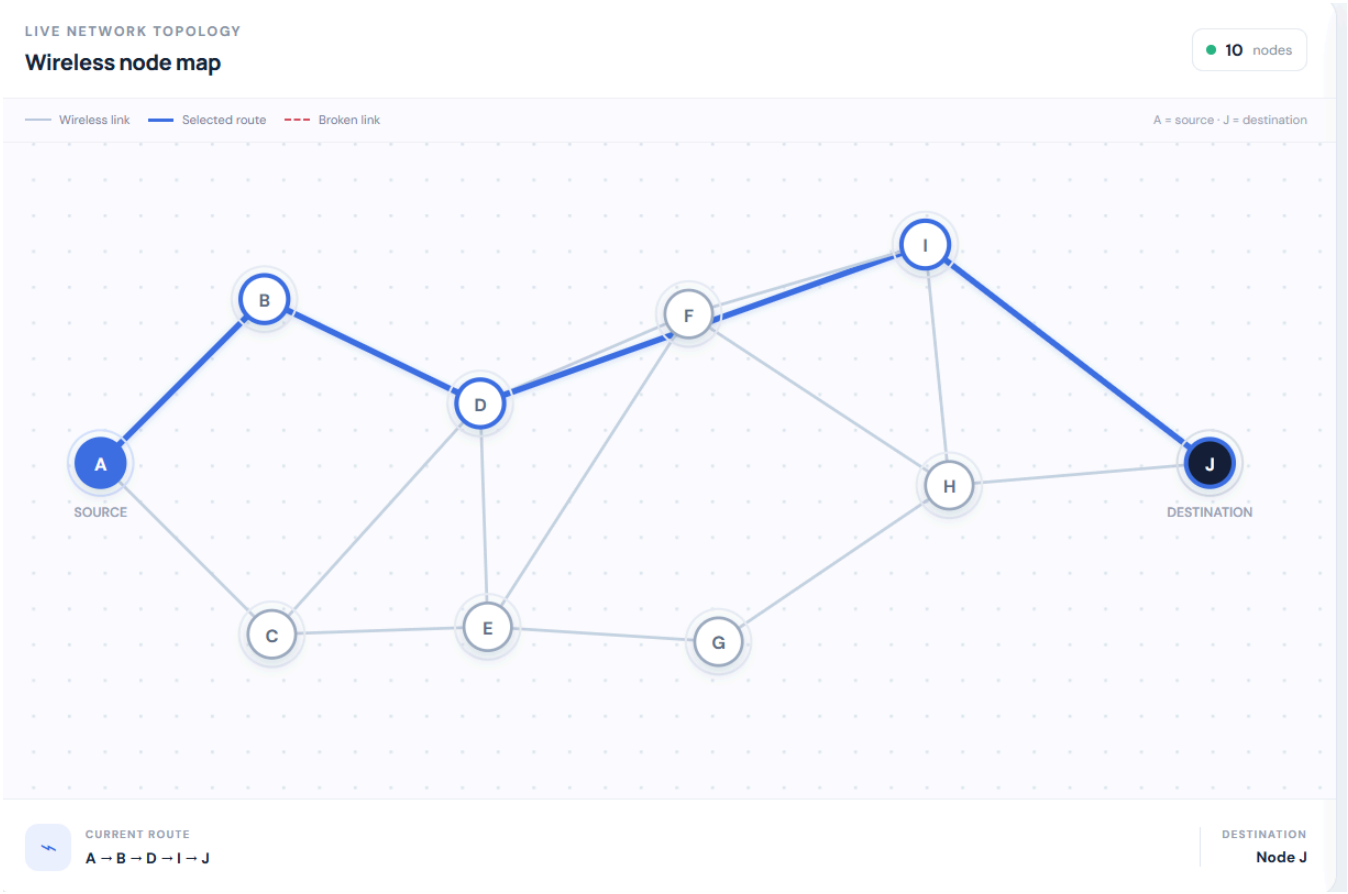
Working process:

1. The source checks whether it has a route.
2. If there is no route, it broadcasts a Route Request (RREQ).
3. Intermediate nodes forward the request.
4. A Route Reply (RREP) returns when a route is found.
5. The source sends data.

6. If a link breaks, a Route Error (RERR) can trigger route recovery or rediscovery.

Advantage: Route discovery is performed when needed.

Limitation: Communication may be delayed while a route is discovered.



4. Mobile IP Application Design for Smart Healthcare (BL4 – Analyze)

Design a Smart Healthcare Mobile IP system using Figma that enables doctors, ambulances, hospitals, and patients to remain connected while moving across different wireless networks.

Analyze:

- How Mobile IP maintains uninterrupted communication.
- Challenges during handoff between networks.
- Security vulnerabilities during mobility.
- Possible design improvements to increase reliability.

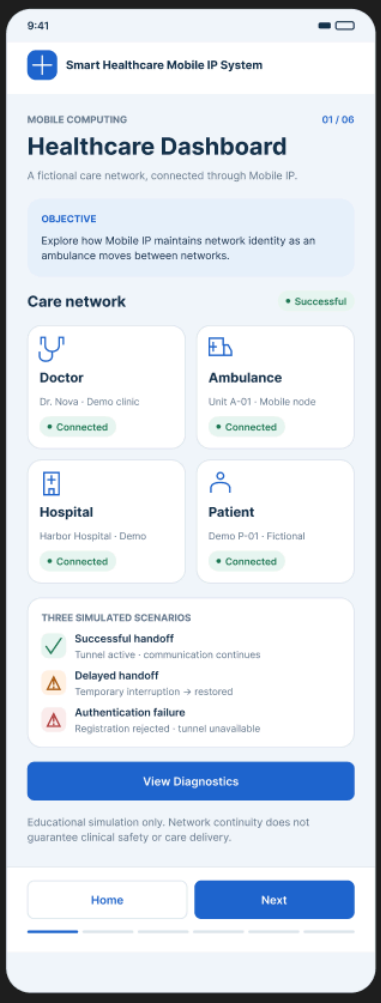
Provide navigation between screens demonstrating different mobility scenarios.

Explanation

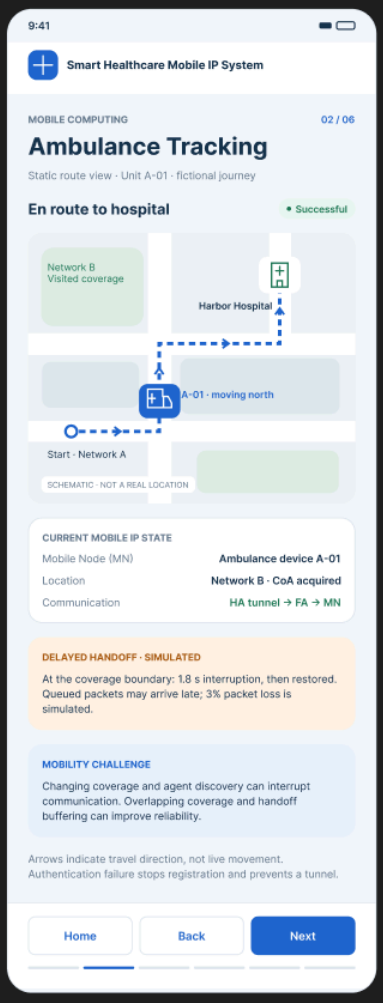
In a smart healthcare system, an ambulance may move through different wireless networks while communicating with a hospital. Mobile IP concepts can support IP-layer communication continuity as a mobile device changes its point of attachment.

Your Figma prototype demonstrates this process through healthcare app screens and simulated network states.

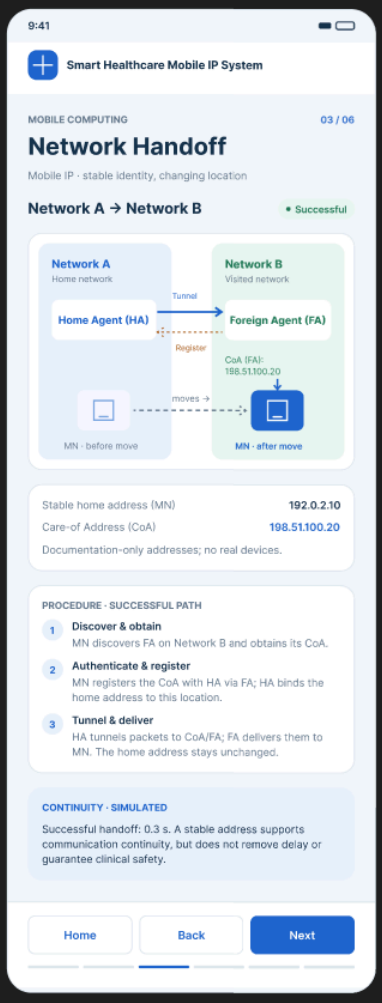
Healthcare Dashboard

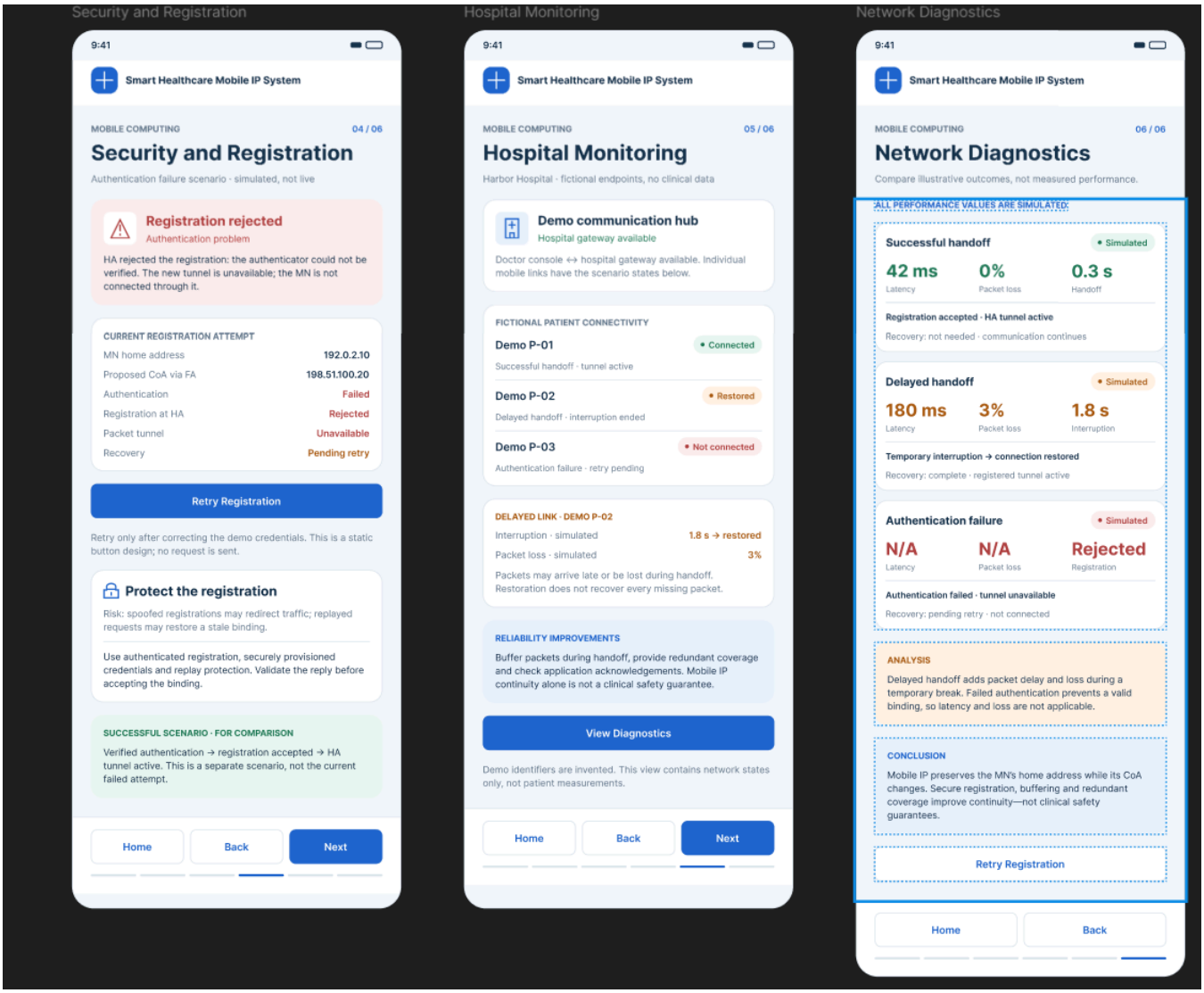


Ambulance Tracking



Network Handoff




Security and Registration
9:41
Smart Healthcare Mobile IP System
MOBILE COMPUTING
04 / 06
Security and Registration
Authentication failure scenario · simulated, not live
Registration rejected
Authentication problem
HA rejected the registration: the authenticator could not be verified. The new tunnel is unavailable; the MN is not connected through it.
CURRENT REGISTRATION ATTEMPT
MN home address
192.0.2.10
Proposed CoA via FA
198.51.100.20
Authentication
Failed
Registration at HA
Rejected
Packet tunnel
Unavailable
Recovery
Pending retry
Retry Registration
Retry only after correcting the demo credentials. This is a static button design; no request is sent.
Protect the registration
Risk: spoofed registrations may redirect traffic; replayed requests may restore a stale binding.
Use authenticated registration, securely provisioned credentials and replay protection. Validate the reply before accepting the binding.
SUCCESSFUL SCENARIO · FOR COMPARISON
Verified authentication → registration accepted → HA tunnel active. This is a separate scenario, not the current failed attempt.
Home
Back
Next
Hospital Monitoring
9:41
Smart Healthcare Mobile IP System
MOBILE COMPUTING
05 / 06
Hospital Monitoring
Harbor Hospital · fictional endpoints, no clinical data
Demo communication hub
Hospital gateway available
Doctor console ↔ hospital gateway available. Individual mobile links have the scenario states below.
FICTIONAL PATIENT CONNECTIVITY
Demo P-01
Connected
Successful handoff · tunnel active
Demo P-02
Restored
Delayed handoff · interruption ended
Demo P-03
Not connected
Authentication failure · retry pending
DELAYED LINK · DEMO P-02
Interruption · simulated
1.8 s → restored
Packet loss · simulated
3%
Packets may arrive late or be lost during handoff. Restoration does not recover every missing packet.
RELIABILITY IMPROVEMENTS
Buffer packets during handoff, provide redundant coverage and check application acknowledgements. Mobile IP continuity alone is not a clinical safety guarantee.
View Diagnostics
Demo identifiers are invented. This view contains network states only, not patient measurements.
Home
Back
Next
Network Diagnostics
9:41
Smart Healthcare Mobile IP System
MOBILE COMPUTING
06 / 06
Network Diagnostics
Compare illustrative outcomes, not measured performance.
ALL PERFORMANCE VALUES ARE SIMULATED.
Successful handoff
Simulated
42 ms
Latency
0%
Packet loss
0.3 s
Handoff
Registration accepted · HA tunnel active
Recovery: not needed · communication continues
Delayed handoff
Simulated
180 ms
Latency
3%
Packet loss
1.8 s
Interruption
Temporary interruption → connection restored
Recovery: complete · registered tunnel active
Authentication failure
Simulated
N/A
Latency
N/A
Packet loss
Rejected
Registration
Authentication failed · tunnel unavailable
Recovery: pending retry · not connected
ANALYSIS
Delayed handoff adds packet delay and loss during a temporary break. Failed authentication prevents a valid binding, so latency and loss are not applicable.
CONCLUSION
Mobile IP preserves the MN's home address while its CoA changes. Secure registration, buffering and redundant coverage improve continuity—not clinical safety guarantees.
Retry Registration
Home
Back

5. Wireless Network Decision Dashboard Using GitHub Project (BL4 – Analyze)

Create a web-based dashboard (using HTML/CSS/JavaScript, React, or another programming framework) or an interactive Figma dashboard that helps network engineers determine whether a given scenario should use Mobile IP, Cellular Networks, or Ad Hoc Networks.

The dashboard should analyze multiple parameters such as:

- Node mobility.
- Infrastructure availability.
- Network size.
- Routing complexity.
- Energy consumption.
- Reliability requirements.

The system should justify its recommendation based on the analyzed parameters.

Explanation

Different wireless networking solutions are suitable for different conditions. A network engineer must consider whether infrastructure is available, how many devices are involved, how frequently they move, and how reliable communication must be.

Your HTML, CSS and JavaScript project collects these requirements and applies decision rules to generate an explanation.

STEP 1

Describe your scenario

Choose the closest values for your network.

Start with an example

Custom scenario

Node mobility

Medium

How frequently do devices move?

Infrastructure availability

Available

Are access networks and services present?

Network size

Large

Routing complexity

High

Energy constraint

Medium

High means battery use is tightly limited.

Reliability requirement

High

Must the device preserve IP communication identity while changing networks?

Yes — IP continuity is required

✦ Analyze scenario

Reset

This demo processes the selections locally in your browser; no data is sent to a server.

STEP 2

Analyzed

Recommendation

IP

BEST MATCH UNDER THESE RULES

Mobile IP mobility support

Use over a suitable underlying wireless/IP access network; Mobile IP does not replace the access network itself.

Why this fits

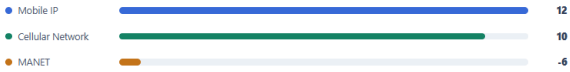
- ✓ Preserving IP communication identity while changing networks is an explicit requirement.
- ✓ Existing IP access networks may provide the underlying connectivity required for Mobile IP.

Trade-offs to consider

- ! Requires compatible mobility support, secure registration, and route/tunnel planning; handoff can still add latency.

Compare suitability scores

Rule score, not probability



This recommendation is generated by simple educational rules. It is not a validated engineering assessment and does not guarantee coverage, performance, or reliability.

Criterion	Marks
Design Quality (Figma/Programming Tool)	20
Technical Analysis	30
Correctness of Mobile Networking Concepts	20
GitHub Repository Organization & Documentation	15
Creativity and Visualization	15
Total	**100 Marks**